

VIGNESH KN

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Professional Summary

Results-driven AI Software Engineer with 4+ years of experience in full-stack engineering, cloud automation, and API integration. Expertise in architecting production-grade Generative AI, LLM orchestration, Agentic workflows, RAG, and NLP pipelines. Skilled in enterprise .NET ecosystems, Sitecore CMS platforms, and secure IAM/PAM infrastructure deployments. Proven track record of designing, scaling, and deploying robust AI and cloud solutions on modern infrastructure.

Technical Skills

AI, GenAI & NLP	Natural Language Processing (NLP), Generative AI (GenAI), Large Language Models (LLMs), Agentic AI Workflows, Prompt Engineering, RLHF Evaluation Frameworks, RAG Architectures
AI Frameworks & Tools	LangChain, LLM APIs (OpenAI, Azure OpenAI), Vector Search (Cosmos DB), FAISS
Languages	C# (.NET Core, ASP.NET MVC), Python, JavaScript, HTML, CSS, SSMS SQL
Frameworks & CMS	Angular (8–19), Entity Framework
Cloud & DevOps	Docker, Azure Functions, Azure Service Bus, Cosmos DB, Git, Jenkins (CI/CD / MLOps), Postman, Swagger

Professional Experience

ACL Digital — Software Engineer	Aug 2024 – Present
% Architected full-stack modules with Agentic AI and LLM APIs for the TLK Device Configuration App (OnePortal) using .NET Core 8, Angular 18, EF, and MySQL globally.	
% Implemented Reinforcement Learning from Human Feedback (RLHF) strategies and developed rigorous evaluation frameworks to optimize the accuracy, performance, and reliability of internal Generative AI and NLP applications.	
% Engineered automated data validation and a secure end-to-end scanner implementation designed specifically for client-side Device Registration.	
Enmarq Technologies — Software Engineer	Aug 2022 – Aug 2024
% Built serverless REST APIs, Azure Functions, and Azure Service Bus triggers utilizing Azure Cosmos DB (Vector Search data management) and SSMS SQL to power low-latency data ingestion pipelines for downstream NLP pipelines and Generative AI RAG architectures.	
% Managed Git and Jenkins CI/CD and MLOps pipelines to ensure seamless version control, automated testing, model deployment, and continuous cloud environment operations.	
% Spearheaded the automated deployment, building, and testing of BeyondTrust PAM and BeyondInsight for 6,000+ users under a strict 90-day timeline.	
% Executed seamless sequential Sitecore CMS upgrades from v8.3 to 10.3, enhancing Content Hub and Experience Editor workflows.	
% Developed complex SQL Stored Procedures for Managed Printing Systems to automate multi-stage data integrity alerts.	
Enmarq Technologies — Intern Associate	Feb 2022 – Jul 2022
% Built automated Python and C# scheduled jobs to extract and clean Cosmos DB user data for analytics and AI workflows.	
% Validated microservices endpoints across backend APIs using Postman and Swagger.	

Certifications & Education

% Microsoft Certified: Azure Fundamentals (AZ-900 / DP-900)	2024 - 2026
% Oracle Cloud Infrastructure (OCI): Generative AI / AI Foundation Certified	2023 - 2024
% B.E. Graduate — KVG College of Engineering	Graduated 2019

Projects & System Architectures (Sourced from GitHub)

VERO — AI Code Analysis & Pull Request Intelligence [SEV-1 · CRITICAL]

<https://github.com/KN-Vignesh/VERO>

Tech Stack: TypeScript, Node.js, GitHub API, SonarQube Rules, TypeSafe Jev, LLMs, Deterministic Policy Code

%I Evidence-based GitHub Pull Request engineering analysis platform combining AST diff parsing, SonarQube static quality checks, and structured LLM signals with a deterministic rule engine.

%I Delivers automated, hallucination-free merge verdicts without repository checkouts, providing verifiable audit trails and structured Markdown review summaries.

Intelligent Customer Churn Prediction System [SEV-2 · HIGH]

<https://github.com/KN-Vignesh/intelligent-customer-churn-prediction>

Tech Stack: Python, Scikit-learn, XGBoost, FastAPI, Docker, Pydantic, Tabular ML

%I End-to-end reproducible classification pipeline transforming raw Telco records into low-latency prediction endpoints.

%I Features isolated Scikit-learn preprocessing pipelines, multi-model evaluation (Logistic Regression, Random Forest, XGBoost), cost-sensitive threshold tuning, and Alpine Docker containerization.

Qwen2.5 / LoRA & QLoRA Memory-Efficient LLM Fine-Tuning [SEV-3 · HIGH]

<https://github.com/KN-Vignesh/Projects/tree/main/Ai-Cookbook/LoraFine-tuning>

Tech Stack: Python, PyTorch, Hugging Face PEFT/TRL, BitsAndBytes 4-bit, LoRA / QLoRA, NF4

%I Parameter-Efficient Fine-Tuning (PEFT) on open-weight LLMs using rank-decomposed adapter matrices (LoRA r=8/16, alpha=32) and 4-bit NormalFloat (NF4) quantization.

%I Preserves base model frozen weights while optimizing attention projection layers (q_proj, v_proj), generating isolated <50MB adapter weights.

BERT Bidirectional Model Engineering & Downstream NLP [SEV-4 · MEDIUM]

https://github.com/KN-Vignesh/Projects/tree/main/Ai-Cookbook/BERT_MODEL

Tech Stack: PyTorch, Hugging Face Transformers, WordPiece Tokenizer, Transfer Learning, AdamW

%I Adapted pretrained bidirectional Transformer encoders to downstream text classification and entity extraction.

%I Engineered pooled CLS linear heads, custom tokenization pipelines, and AdamW linear learning rate warmup with gradient clipping (1.0).

Multi-Metric Model Evaluation & Experiment Benchmarking Suite [SEV-5 · MEDIUM]

https://github.com/KN-Vignesh/Projects/tree/main/Ai-Cookbook/Combined_metric_Calc

Tech Stack: Python, Scikit-learn, ROC-AUC, Precision-Recall, Calibration Curves, Bootstrap

%I Unified model comparison suite synthesizing discrimination metrics, calibration curves (Brier score), and latency metrics across experiments for evidence-based deployment justification.